

| Question     |                   |              | Marking details                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |                   |              |       |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
|--------------|-------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|--------------|-------|--------|------|------|-----|------|------|---------|------|-------|------|---------|--|---|--|---|---|---|
|              |                   |              |                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2               | AO3          | Total | Maths  | Prac |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
| 5            | (a)               |              | For a system to be in equilibrium (1)<br>$\sum$ anticlockwise moments [about a point] = $\sum$ clockwise moments [about the same point] (1)<br>N.B. Award 1 mark for $\sum$ C.M = $\sum$ A.C.M. only<br><b>Alternative:</b><br>For a system to be in equilibrium (1) algebraic sum of moments / net moment / resultant moment [about a point] = 0 (1)                                                  | 2               |                   |              | 2     |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
|              | (b)               |              | Centre of gravity or line of action of weight remained inside lower surface of block [or equiv, e.g. CoG remains to left of pivot]<br>Accept arrow clearly shown on diagram [1]<br>Produces anticlockwise <u>moment</u> (or torque) [about pivot] [1]                                                                                                                                                  |                 | 2                 |              | 2     |        | 2    |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
|              | (c)               | (i)          | All mean $F$ values determined correctly [1]<br><table><tr><td>Mean <math>F</math> / N</td><td rowspan="7"><b>Or accept:</b></td><td>Mean <math>F</math> / N</td></tr><tr><td>2.7</td><td>2.7[0]</td></tr><tr><td>5.5</td><td>5.45</td></tr><tr><td>8.7</td><td>8.65</td></tr><tr><td>11.4</td><td>11.4[0]</td></tr><tr><td>14.6</td><td>14.55</td></tr><tr><td>17.4</td><td>17.4[0]</td></tr></table> | Mean $F$ / N    | <b>Or accept:</b> | Mean $F$ / N | 2.7   | 2.7[0] | 5.5  | 5.45 | 8.7 | 8.65 | 11.4 | 11.4[0] | 14.6 | 14.55 | 17.4 | 17.4[0] |  | 1 |  | 1 | 1 | 1 |
| Mean $F$ / N | <b>Or accept:</b> | Mean $F$ / N |                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                   |              |       |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
| 2.7          |                   | 2.7[0]       |                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                   |              |       |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
| 5.5          |                   | 5.45         |                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                   |              |       |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
| 8.7          |                   | 8.65         |                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                   |              |       |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
| 11.4         |                   | 11.4[0]      |                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                   |              |       |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
| 14.6         |                   | 14.55        |                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                   |              |       |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |
| 17.4         |                   | 17.4[0]      |                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                   |              |       |        |      |      |     |      |      |         |      |       |      |         |  |   |  |   |   |   |